WORKED SOLUTION

STEP II 2021, Question 1

Pure •

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Products of sines and cosines are awkward, but sums are easy to handle, so the opening identities let us trade a product for a sum. In (i) that turns a triple product into a cubic in $\cos 2x$ with a visible factor. In (ii) we clear the tangents, pair each factor with its partner of the opposite type, and the whole equation collapses into a product of two simple factors. The only extra care needed is over the points where a tangent is undefined.

Solution

The identities

Adding the expansions

$$\cos(A+B) = \cos A\cos B - \sin A\sin B, \qquad \cos(A-B) = \cos A\cos B + \sin A\sin B$$

gives $\cos(A+B)+\cos(A-B) = 2\cos A\cos B$. Taking $A = 3a$ and $B = a$,

$$\cos a\cos 3a = \tfrac{1}{2}(\cos 4a + \cos 2a),$$

as required. In the same way, adding $\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B$ gives $\sin(A+B)+\sin(A-B) = 2\sin A\cos B$. With $A = a$ and $B = 3a$, and using $\sin(-2a) = -\sin 2a$,

$$\sin a\cos 3a = \tfrac{1}{2}(\sin 4a - \sin 2a).$$

(Swapping the roles, $A = 3a$ and $B = a$, gives the companion $\cos a\sin 3a = \tfrac{1}{2}(\sin 4a + \sin 2a)$, which we shall also need.)

Part (i)

By the first identity, $4\cos x\cos 2x\cos 3x = 2\cos 2x\,(\cos 4x + \cos 2x)$. Write $c = \cos 2x$, so that $\cos 4x = 2c^2 - 1$. The equation becomes

$$2c\,(2c^2 + c - 1) = 1 \quad\Longleftrightarrow\quad 4c^3 + 2c^2 - 2c - 1 = 0.$$

Grouping the terms in pairs, $2c^2(2c+1) - (2c+1) = 0$, that is,

$$(2c+1)(2c^2-1) = 0,$$

so $\cos 2x = -\frac{1}{2}$ or $\cos 2x = \pm\frac{1}{\sqrt{2}}$.

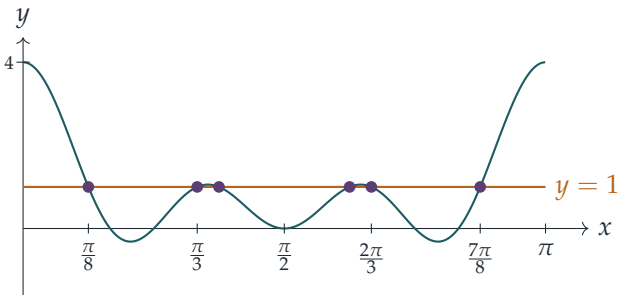
For $0 \le x \le \pi$ we have $0 \le 2x \le 2\pi$, and each value of $\cos 2x$ strictly between -1 and 1 gives two values of $2x$ in that range:

$$\begin{aligned}
\cos 2x = -\tfrac{1}{2} &\implies 2x = \tfrac{2\pi}{3},\ \tfrac{4\pi}{3} &&\implies x = \tfrac{\pi}{3},\ \tfrac{2\pi}{3};\\
\cos 2x = \tfrac{1}{\sqrt{2}} &\implies 2x = \tfrac{\pi}{4},\ \tfrac{7\pi}{4} &&\implies x = \tfrac{\pi}{8},\ \tfrac{7\pi}{8};\\
\cos 2x = -\tfrac{1}{\sqrt{2}} &\implies 2x = \tfrac{3\pi}{4},\ \tfrac{5\pi}{4} &&\implies x = \tfrac{3\pi}{8},\ \tfrac{5\pi}{8}.
\end{aligned}$$

Every step was reversible, so the solutions are

$$x = \frac{\pi}{8},\ \frac{\pi}{3},\ \frac{3\pi}{8},\ \frac{5\pi}{8},\ \frac{2\pi}{3},\ \frac{7\pi}{8}.$$

The graph confirms there are exactly six crossings of the line $y = 1$.



Part (ii)

Suppose $\tan x = \tan 2x \tan 3x \tan 4x$. For this even to make sense, all four tangents must be defined, so $\cos x$, $\cos 2x$, $\cos 3x$ and $\cos 4x$ are all non-zero. We may therefore multiply through by $\cos x \cos 2x \cos 3x \cos 4x$ to get

$$\sin x \cos 3x \cdot \cos 2x \cos 4x = \cos x \sin 3x \cdot \sin 2x \sin 4x.$$

? **Ask yourself** Why pair $\sin x$ with $\cos 3x$, and $\cos x$ with $\sin 3x$?

Because those are exactly the products our identities convert into sums, and both sums involve only $\sin 4x$ and $\sin 2x$, which already appear elsewhere in the equation. Using $\sin x \cos 3x = \frac{1}{2}(\sin 4x - \sin 2x)$ and $\cos x \sin 3x = \frac{1}{2}(\sin 4x + \sin 2x)$ and doubling,

$$(\sin 4x - \sin 2x)\cos 2x \cos 4x = (\sin 4x + \sin 2x)\sin 2x \sin 4x.$$

Now replace $\sin 2x \cos 2x$ by $\frac{1}{2}\sin 4x$ on the left, so that every term has a factor $\sin 4x$:

$$\sin 4x \cos 4x \cos 2x - \tfrac{1}{2}\sin 4x \cos 4x = \sin 4x \sin^2 2x + \sin 4x \sin 4x \sin 2x.$$

Taking everything to one side,

$$\sin 4x\left[(\cos 4x\cos 2x-\sin 4x\sin 2x)-\left(\sin^2 2x+\tfrac{1}{2}\cos 4x\right)\right]=0.$$

The first bracket is $\cos(4x+2x)=\cos 6x$. For the second, $\sin^2 2x=\frac{1}{2}(1-\cos 4x)$, so $\sin^2 2x+\frac{1}{2}\cos 4x=\frac{1}{2}$. Hence

$$\sin 4x\left(\cos 6x-\tfrac{1}{2}\right)=0,$$

so $\cos 6x=\frac{1}{2}$ or $\sin 4x=0$, as required.

Now for the solutions of $(\star)$ with $0\le x\le\pi$.

Ask yourself Is every solution of $\sin 4x(\cos 6x-\frac{1}{2})=0$ a solution of $(\star)$?

Not necessarily: the only step that could fail to reverse was multiplying by the cosines, and that is reversible precisely when all four tangents are defined. So the solutions of $(\star)$ are the solutions of $\sin 4x(\cos 6x-\frac{1}{2})=0$ at which $\cos x,\cos 2x,\cos 3x,\cos 4x\neq 0$.

Case $\sin 4x=0$. In $[0,\pi]$ this gives $x=0,\ \frac{\pi}{4},\ \frac{\pi}{2},\ \frac{3\pi}{4},\ \pi$. At $x=\frac{\pi}{4}$ and $x=\frac{3\pi}{4}$, $\tan 2x$ is undefined; at $x=\frac{\pi}{2}$, $\tan x$ is undefined. At $x=0$ and $x=\pi$ every tangent is 0, and $(\star)$ reads $0=0$. So $x=0$ and $x=\pi$ are solutions.

Case $\cos 6x=\frac{1}{2}$. Then $6x=\pm\frac{\pi}{3}+2k\pi$, so $x=\frac{(6k\pm1)\pi}{18}$. With $0\le 6x\le 6\pi$ this gives

$$x=\frac{\pi}{18},\ \frac{5\pi}{18},\ \frac{7\pi}{18},\ \frac{11\pi}{18},\ \frac{13\pi}{18},\ \frac{17\pi}{18}.$$

Each is $\frac{m\pi}{18}$ with m not a multiple of 3. A cosine $\cos nx$ vanishes only when nx is an odd multiple of $\frac{\pi}{2}=\frac{9\pi}{18}$, which would need $9\mid nm$. Since $3\nmid m$, this would need $9\mid n$, which is impossible for $n\in\{1,2,3,4\}$. So all four tangents are defined and these six values are all solutions.

Altogether $(\star)$ has eight solutions in $0\le x\le\pi$:

$$x=0,\ \frac{\pi}{18},\ \frac{5\pi}{18},\ \frac{7\pi}{18},\ \frac{11\pi}{18},\ \frac{13\pi}{18},\ \frac{17\pi}{18},\ \pi.$$

Checks

- In (i), at $x=\frac{\pi}{8}$: $\cos\frac{\pi}{4}=\frac{1}{\sqrt{2}}$ and $\cos\frac{3\pi}{8}\cos\frac{\pi}{8}=\frac{1}{2}(\cos\frac{\pi}{2}+\cos\frac{\pi}{4})=\frac{1}{2\sqrt{2}}$, so the product is $4\cdot\frac{1}{\sqrt{2}}\cdot\frac{1}{2\sqrt{2}}=1$.
- In (ii), at $x=\frac{\pi}{18}=10^\circ$: $\tan 10^\circ\approx 0.17633$, while $\tan 20^\circ\tan 30^\circ\tan 40^\circ\approx 0.36397\times 0.57735\times 0.83910\approx 0.17633$.
- A computer scan of both equations over $[0,\pi]$ finds exactly six and eight solutions respectively, at the values above.

Common slips

- Keeping $x = \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}$ in (ii): they satisfy $\sin 4x = 0$ but make a tangent in $(\star)$ undefined.
- Forgetting $x = 0$ and $x = \pi$, or missing the second value of $2x$ (or $6x$) in each period.
- Dividing by $\sin 4x$ during the simplification, which loses the factor that the question wants you to find.

Beyond STEP

The product-to-sum formulae are the real-variable shadow of $e^{iA}e^{iB} = e^{i(A+B)}$: writing $\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$ turns any product of cosines into a sum, which is the first step towards Fourier series, where functions are expanded as sums of $\cos nx$ and $\sin nx$. The striking identity $\tan 10^\circ = \tan 20^\circ \tan 30^\circ \tan 40^\circ$ hidden in (ii) is typical of the surprises that appear when trigonometric values are viewed as roots of unity. Needham's *Visual Complex Analysis* develops this viewpoint beautifully.

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